# EDA

## Introduction

The data being used in this project is sourced from the Behavioral Risk Factor Surveillance System (BRFSS) which is a telephone based survey that is collected annually be the CDC. There are three data sets available from the BRFSS, but the scope of this analysis is limited to one, diabetes _ binary _ health _ indicators _ BRFSS2015.csv. This is a clean data set with 253,680 responses. The variable of interest is Diabetes_binary which denotes no diabetes with 0 and pre-diabetes or diabetes with 1. There are 21 other variables included that may have relation to the variable of interest.

The variables that will be used in EDA and modelling are BMI, whether or not the responder has higher cholesterol, and whether or not the responder consumes fruit 1 or more times per day. According to the NIH, there is elevated risk of having diabetes when the patient has a higher BMI (https://pmc.ncbi.nlm.nih.gov/articles/PMC4457375/). Diabetes is also known to elevated the "bad" cholesterol in the body while decreasing the amount of "good" cholesterol (https://www.heart.org/en/health-topics/diabetes/diabetes-complications-and-risks/cholesterol-abnormalities–diabetes). Fruit is a naturally sugary food, therefore, many people with diabetes may have to take into account the portions of fruit being eaten which could indicate the presence of diabetes in the responders.

## Data

```r
data <- read.csv("diabetes_binary_health_indicators_BRFSS2015.csv", fileEncoding = "latin1")
```

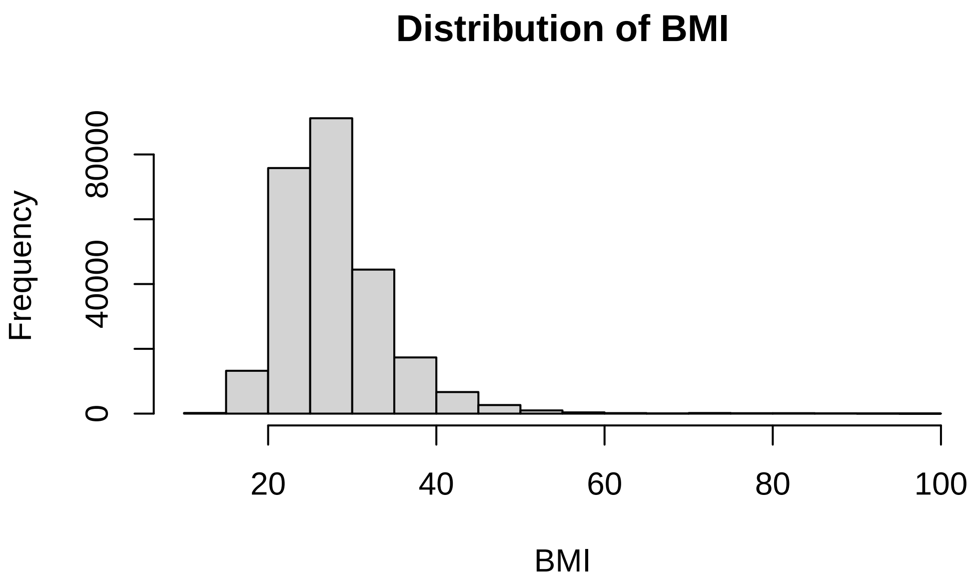
## Summarizations

### Variable 1: BMI

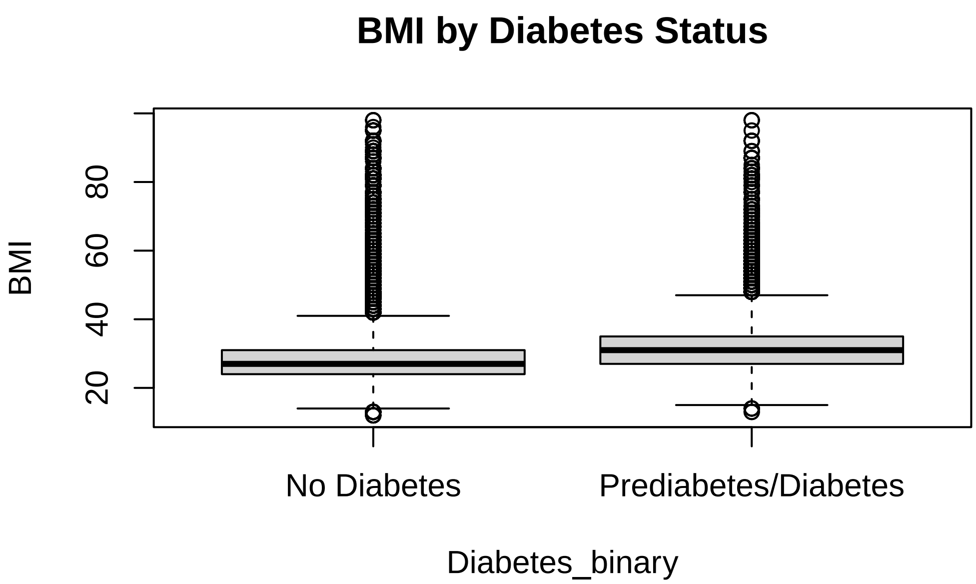
```r
# Univariate
summary(data$BMI)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  12.00   24.00   27.00   28.38   31.00   98.00 
```

```r
hist(data$BMI, breaks = 30, main = "Distribution of BMI", xlab = "BMI")
```




```r
# Bivariate
boxplot(BMI ~ Diabetes_binary, data = data,
        names = c("No Diabetes", "Prediabetes/Diabetes"),
        main = "BMI by Diabetes Status", ylab = "BMI")
```



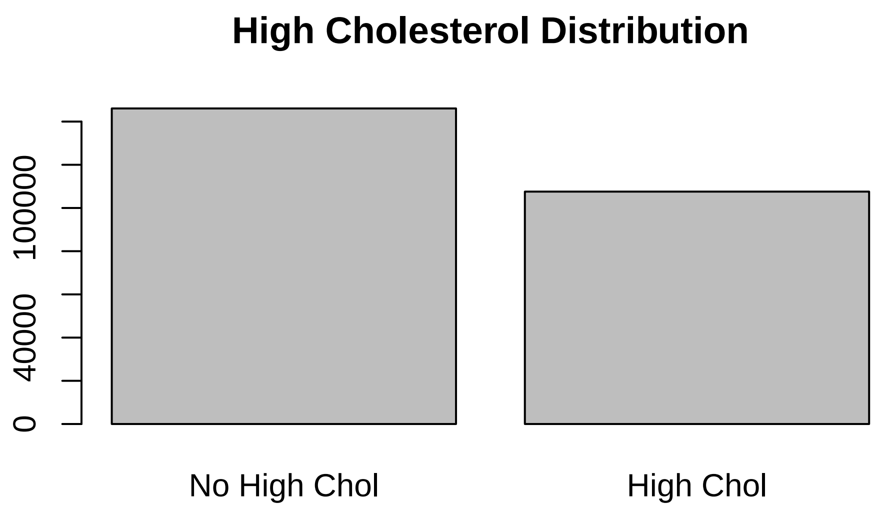

The univariate histogram and summary statistics show that the BMI variable is right skewed with an average of about 28. This makes sense as there is a lower bound for BMI at 0 with a hypothetically infinite upper bound. The bivariate summary using boxplots shows that there is not a significant difference in BMI between the groups with and without diabetes.

#### Variable 2: High Cholesterol

```
# Univariate
prop.table(table(data$HighChol))
```

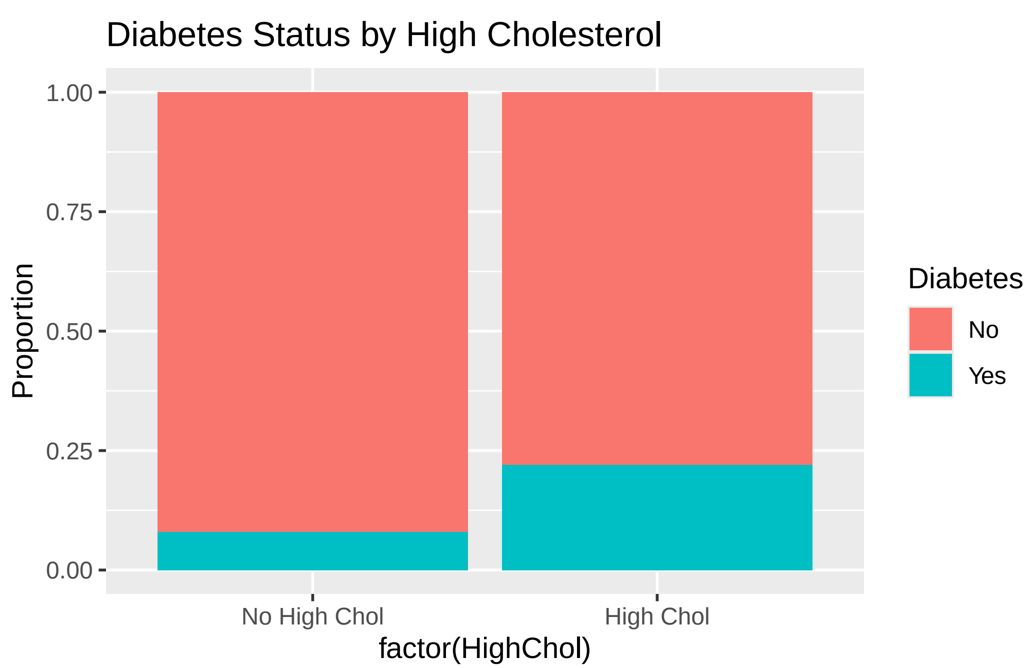
```
        0         1 
0.5758791 0.4241209 
```

```
barplot(table(data$HighChol),
        names.arg = c("No High Chol", "High Chol"),
        main = "High Cholesterol Distribution")
```




```
#Bivariate
library(ggplot2)

ggplot(data, aes(x = factor(HighChol), fill = factor(Diabetes_binary))) +
  geom_bar(position = "fill") +
  scale_x_discrete(labels = c("No High Chol", "High Chol")) +
  scale_fill_discrete(name = "Diabetes", labels = c("No", "Yes")) +
  ylab("Proportion") +
  ggtitle("Diabetes Status by High Cholesterol")
```



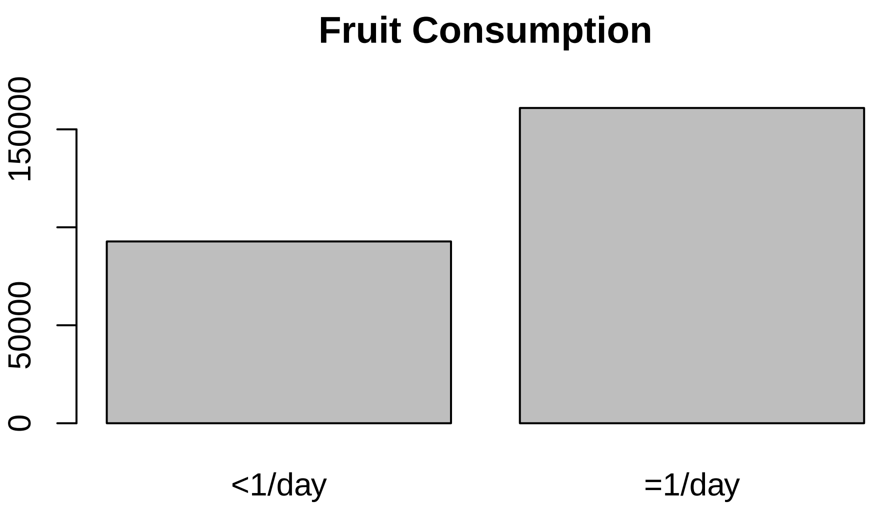
The univariate proportion table and barplot show that there is a larger proportion of reponders without high cholesterol. The difference in proportions between responders with and without high cholesterol is about 0.16. The bivariate barplot with grouping based on whether the responder has diabetes or not shows that there is a larger proportion of responders that have high cholesterol and diabetes than no high cholesterol and diabetes.

#### Variable 3: Fruit Consumed

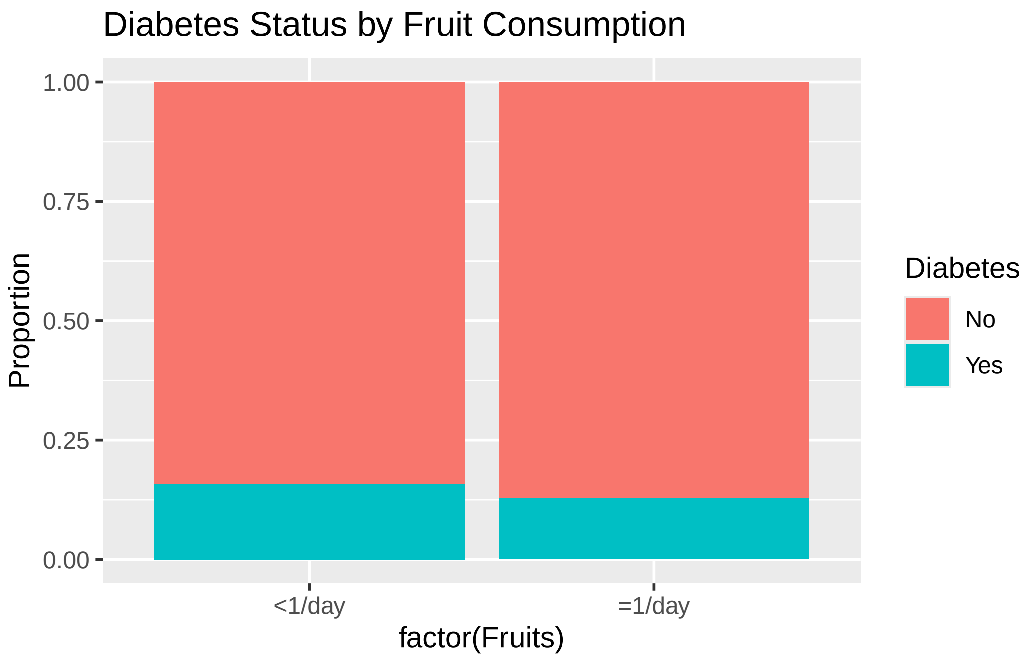
```
# Univariate
prop.table(table(data$Fruits))
```

```
        0         1
0.3657442 0.6342558
```

```
barplot(table(data$Fruits),
        names.arg = c("<1/day", " 1/day"),
        main = "Fruit Consumption")
```




```
# Bivariate
ggplot(data, aes(x = factor(Fruits), fill = factor(Diabetes_binary))) +
  geom_bar(position = "fill") +
  scale_x_discrete(labels = c("<1/day", " 1/day")) +
  scale_fill_discrete(name = "Diabetes", labels = c("No", "Yes")) +
  ylab("Proportion") +
  ggtitle("Diabetes Status by Fruit Consumption")
```




The univariate proportion table and barplot show that there is a higher proportion of responders that eat at least one fruit per day. The bivariate barplot shows that there is not much difference between the proportions of people with and without diabetes and the whether they eat fruit in a day.